

Sorbonne AD

Day 1: Data Science - Project Course

STATISTICS + MACHINE LEARNING + DATA SCIENCE

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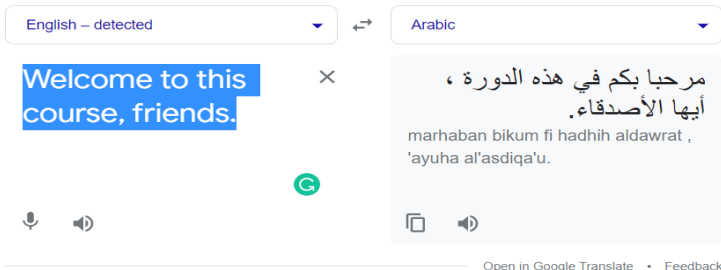
Course for BSc Mathematics and Data Science Students

Why this course?



Let data drive decisions, not the **Highest Paid Person's Opinion**.

Data Science can learn to translate text in the domain of
Natural Language Processing



The screenshot shows the Google Translate web interface. On the left, the 'English - detected' dropdown is selected, and the text 'Welcome to this course, friends.' is entered. On the right, the 'Arabic' dropdown is selected, and the translated text is displayed in Arabic script: 'مرحبا بكم في هذه الدورة ، أيتها الأصدقاء.' Below the Arabic text, a phonetic transcription is provided: 'marhaban bikum fi hadhih aldawrat , 'ayuha al'asdiqa'u.' The interface includes a microphone icon, a speaker icon, a copy icon, and a feedback link at the bottom right.

English – detected

Arabic

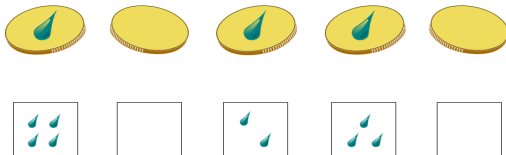
Welcome to this course, friends.

مرحبا بكم في هذه الدورة ،
أيتها الأصدقاء.

marhaban bikum fi hadhih aldawrat ,
'ayuha al'asdiqa'u.

Open in Google Translate • Feedback

Data Science can help us to forecast whether it will rain tomorrow or not (not by tossing a coin)



Tlaloc is the god of rain in Aztec religion

kaggle COMPETITIONS



“Data is the new oil. It’s valuable, but if unrefined it cannot really be used. It has to be changed into gas, plastic, chemicals, etc to create a valuable entity that drives profitable activity; so must data be broken down, analyzed for it to have value.”

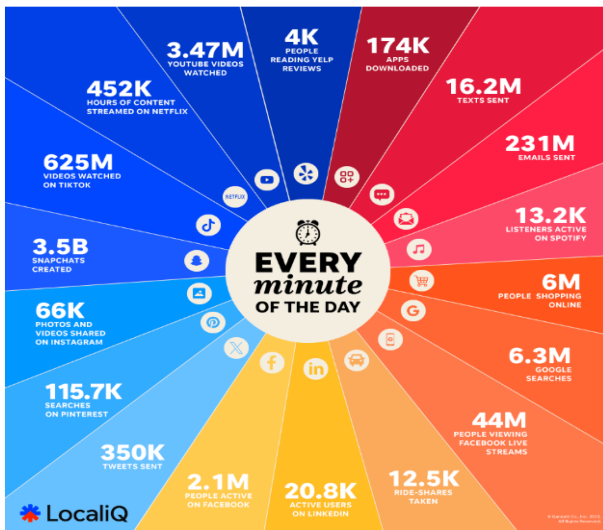
- Clive Humby, UK Mathematician and Architect of Tesco’s Clubcard.

Example:

City	Morning temperature	Evening temperature
Austin	62	90.7
Boston	41	48.0
Chicago	51	57.2
Denver	45	52.5



Role of data: Present



Astronomy



Social Networks



Healthcare



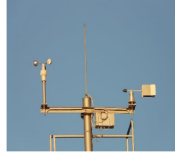
Banking



Genomics

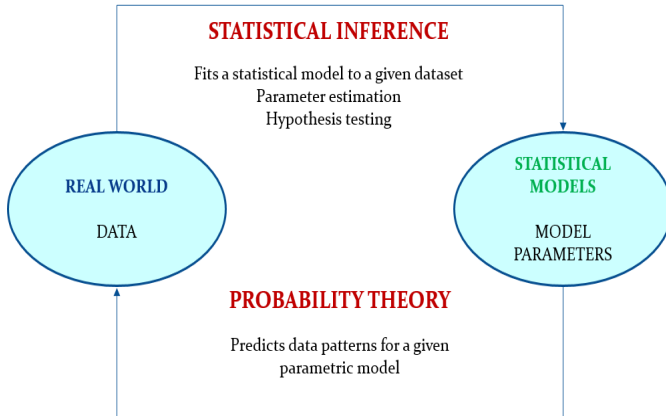


Weather
measurements



- 1 **Statistics** is the study of the collection, analysis, interpretation, presentation and organization of data.
- 2 **Data science** is the study of the generalizable extraction of knowledge from data, yet the key word is science.
- 3 **Machine learning** is the sub-field of computer science that gives computers the ability to learn without being explicitly programmed.
- 4 **Artificial Intelligence** research is defined as the study of intelligent agents: any device that perceives its environment and takes actions that maximize its chance of success at some goal.

TOPIC 1 : STATISTICS



- **Role of Statistics:**
 - 1 Making inference from samples
 - 2 Development of new methods for complex data sets
 - 3 Quantification of uncertainty and variability

- **Two Views of Statistics:**
 - 1 Statistics as a Mathematical Science
 - 2 Statistics as a Data Science

- Data characteristics:
 - Size
 - Dimensionality
 - Complexity
 - Messy
 - Secondary sources
- Focus on generalization performance :
 - Prediction on new data
 - Action in new circumstances
 - Complex models needed for good generalization
- Computational considerations :
 - Large scale and complex systems

TOPIC 2 : DATA SCIENCE

Types of Data Science?

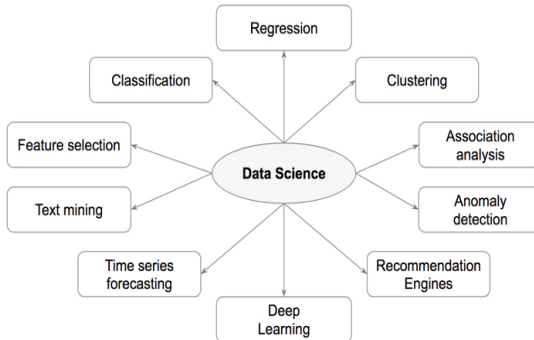
*"When you're fundraising, it's **AI**.*

*When you're hiring, it's **ML**.*

*When you're implementing, it's **Linear Regression**.*

*When you're debugging, it's **printf()**."*

- Baron Schwartz, Founder and CEO of VividCortex, 2017.



Basic Definitions:

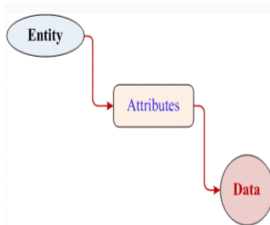
Entity: A particular thing is called entity or object.

Attribute: An attribute is a measurable or observable property of an entity.

Data: A measurement of an attribute is called data.

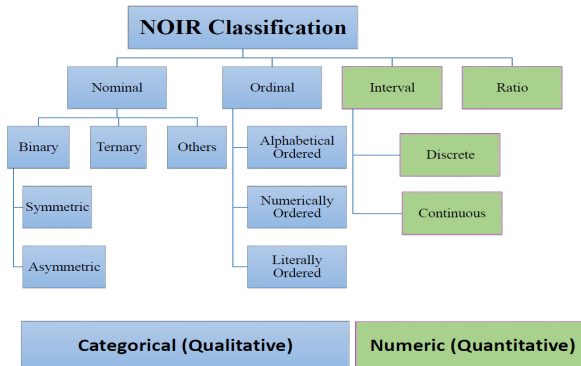
Note: Data defines an entity and Computer can manage all type of data (e.g., audio, video, text, etc.). In general, there are many types of data that can be used to measure the properties of an entity.

Scale: A good understanding of data scales (also called scales of measurement) is important. Depending on the scales of measurement, different techniques are followed to derive hitherto unknown knowledge in the form of patterns, associations, anomalies or similarities from a volume of data.



NAME	AGE	GENDER	SALARY	EMPLOYER
:				
:				
ABCD	34	F	40000	XYZ
:				
:				

- The **NOIR scale** is the fundamental building block on which the extended data types are built.
- Further, nominal (Blood groups, Attendance) and ordinal (Shirt size) are collectively referred to as **categorical or qualitative data**. Whereas, interval (weight, temperature) and ratio (Sound intensity in Decibel) data are collectively referred to as **quantitative or numeric data**.



Concept of data cube:

A **multidimensional data** model views data in the form of a cube. A data cube is characterized with two things:

- **Dimension:** The perspective or entities with respect to which an organization wants to keep record.
- **Fact:** The actual values in the record.

Example: Rainfall data of Meteorological Department.

- Time (Year, Season, Month, Week, Day, etc.)
- Location (Country, Region, State, etc.)

2-D view of rainfall data

- In this 2-D representation, the rainfall for “North-East” region are shown with respect to different months for a period of years...

Region: North-East

	Month											
	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Year	2005											
	2006											
	2007											
	2008											
	2009											
	2010											

- Suppose, we want to represent data according to times (Year, Month) as well as regions of a country say East, West, North, North-East, etc.

East		Month											
Year		Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
	2005												
	2006												
	2007												
	2008												
	2009												
	2010												

Figure: 2-D view of rainfall data

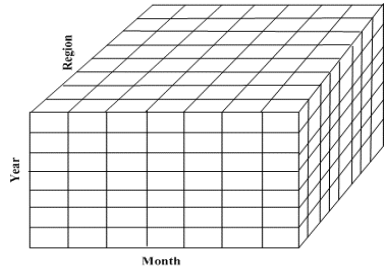
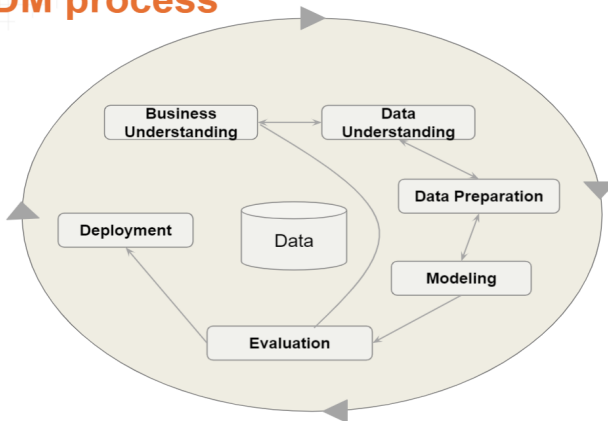
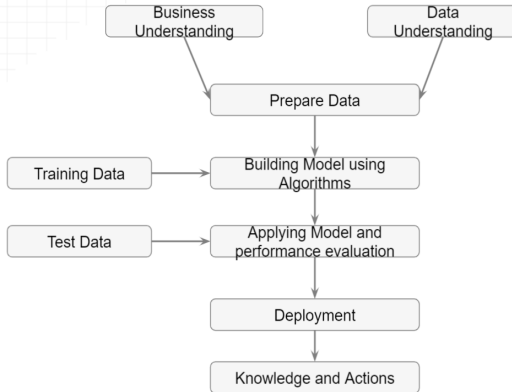


Figure: 3-D view of rainfall data

DM process



Process



1. Prior Knowledge

2. Preparation

3. Modeling

4. Application

5. Knowledge

1. Prior Knowledge

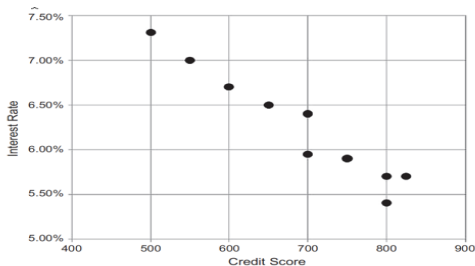
Gaining information on

- Objective of the problem.
- Subject area of the problem.
- Data.

Table 2.1 Data Set		
Borrower ID	Credit Score	Interest Rate
01	500	7.31%
02	600	6.70%
03	700	5.95%
04	700	6.40%
05	800	5.40%
06	800	5.70%
07	750	5.90%
08	550	7.00%
09	650	6.50%
10	825	5.70%

Gaining information on

- Data Exploration and Data quality.
- Handling missing values and Outliers.
- Data type conversion.
- Transformation, Feature selection and Sampling.



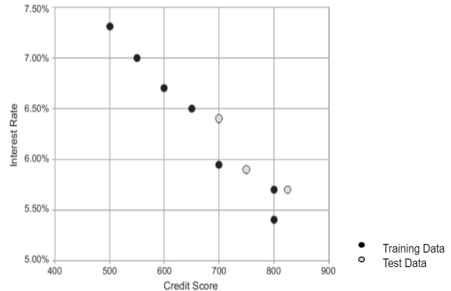
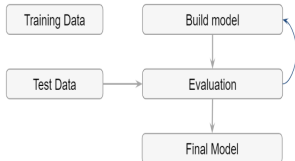
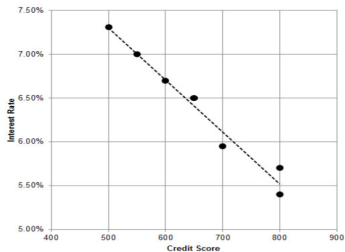


Figure: Splitting data into training and test data sets (right).

3. Modeling



$$y = 0.1 + \frac{6}{100,000}x$$

Table 2.5 Evaluation of Test Data Set

Borrower	Credit Score (X)	Interest Rate (Y)	Model Predicted (Y)	Model Error
04	700	6.40%	6.11%	-0.29%
07	750	5.90%	5.81%	-0.09%
10	825	5.70%	5.37%	-0.33%

Figure: Evaluation of test dataset (right).

4. Application:

- Product readiness.
- Technical integration.
- Model response time.
- Remodeling.
- Assimilation.

5. Knowledge:

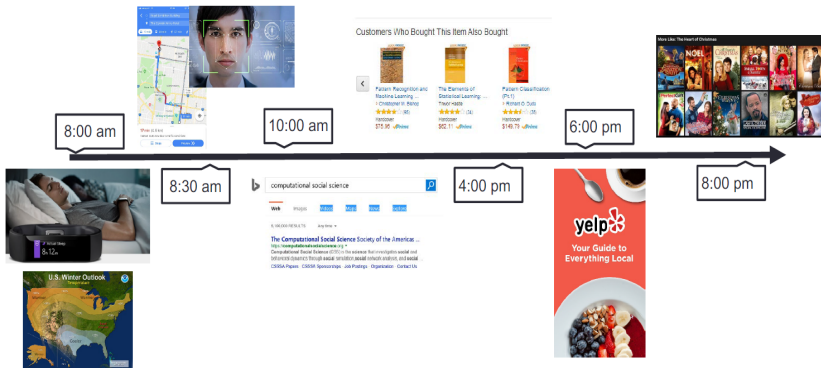
- Posterior knowledge.

Objectives of Data Exploration:

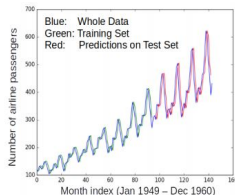
- Understanding data.
- Data preparation and Data mining tasks.
- Interpreting data mining results.

TOPIC 3 : MACHINE LEARNING

A day in our life with ML techniques...

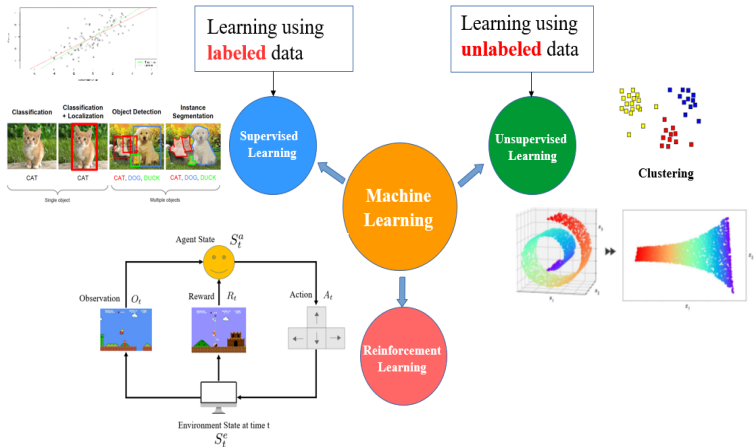


- Designing algorithms that **ingest data** and **learn a model** of the data.
- The learned model can be used to
 - 1 Detect **patterns/structures/themes/trends** etc. in the data
 - 2 Make **predictions** about future data and make decisions

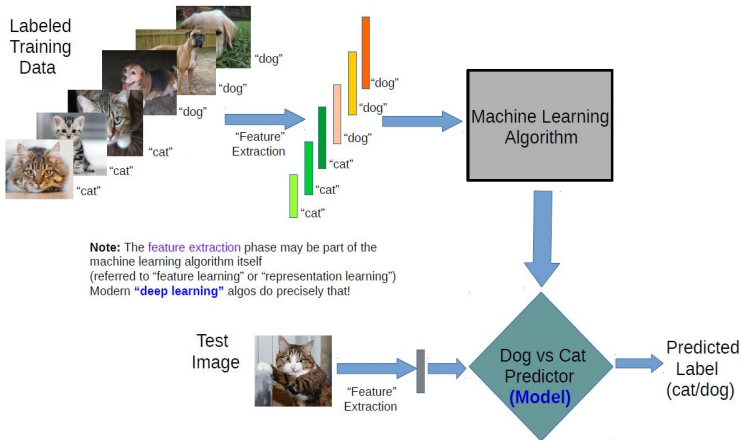


- Modern ML algorithms are heavily **"data-driven"**.
- Optimize a performance criterion using example data or **past experience**.

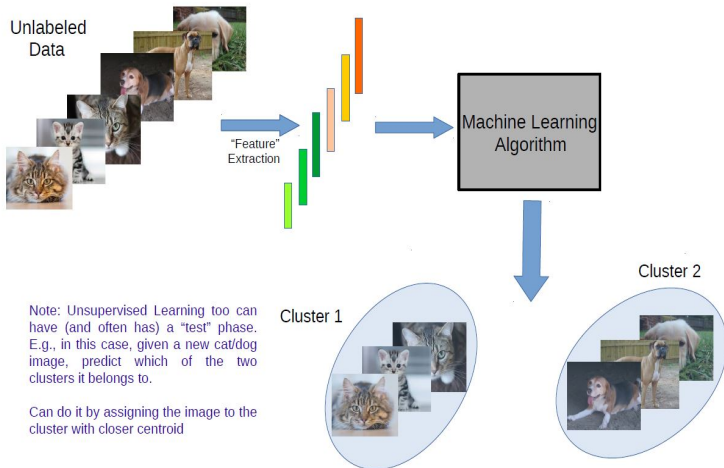
A Loose Taxonomy of ML



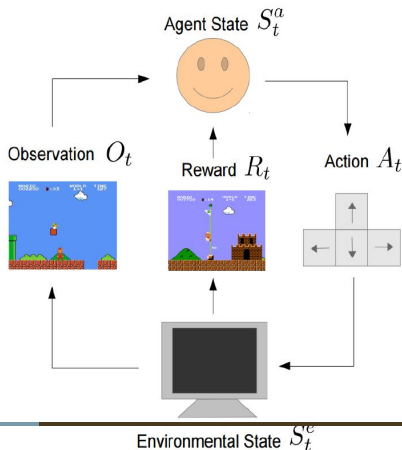
Supervised Learning: Predicting patterns in the data



Unsupervised Learning: Discovering patterns in the data



Reinforcement Learning: Learning a "policy" by performing actions and getting rewards (e.g, robot controls, beating games)



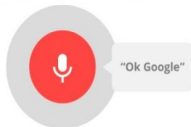
Agent's goal is to learn a policy for some task

Agent does the following repeatedly

- Senses/observes the environment
- Takes an action based on its current policy
- Receives a reward for that action
- Updates its policy

There IS supervision, not explicit (as in Supervised Learning) but rather implicit (feedback based)

Broadly applicable in many domains (e.g., internet, robotics, healthcare and biology, computer vision, NLP, databases, computer systems, finance, etc.).

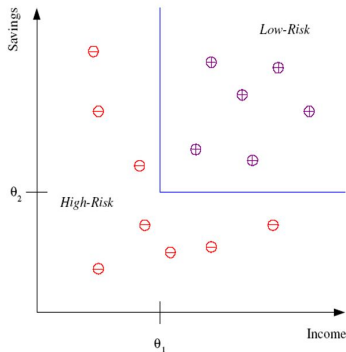


Predictive Policing

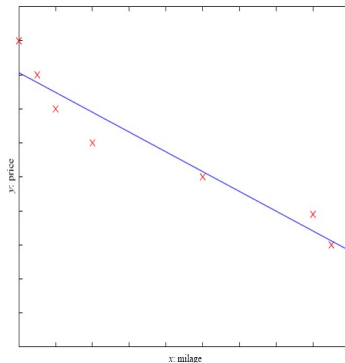


Online Fraud Detection

- **Example:** Credit scoring.
- Differentiating between low-risk and high-risk customers from their income and savings.
- **Discriminant:** IF $\text{Income} > \theta_1$ AND $\text{Savings} > \theta_2$ THEN low-risk ELSE high-risk.
- **Classification:** Learn a linear/nonlinear separator (the "model") using training data consisting of input-output pairs (each output is discrete-valued "label" of the corresponding input).
- Use it to predict the labels for new "test" inputs.
- **Other Applications:** Image Recognition, Spam Detection, Medical Diagnosis.



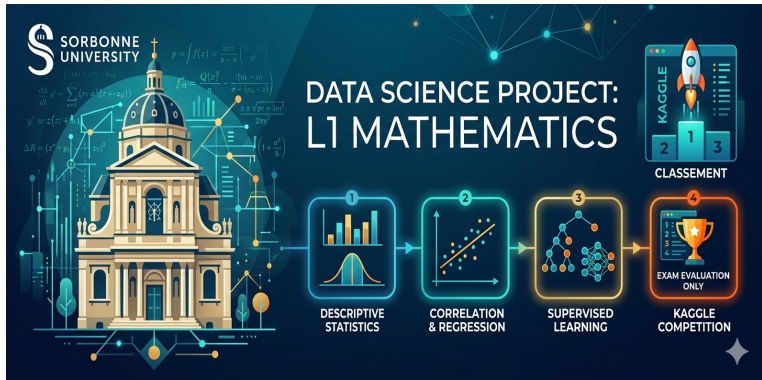
- **Example:** Price of a used car.
- X : car attributes; Y : price and $Y = f(X, \theta)$
- $f()$ is the model and θ is the model parameters.
- **Regression:** Learn a line/curve (the "model") using training data consisting of Input-output pairs (each output is a real-valued number).
- Use it to predict the outputs for new "test" inputs.
- **Other Applications:** Price Estimation, Process Improvement, Weather Forecasting.



What type of data are involved in the following applications?

- Weather forecasting
- Mobile usage of all customers of a service provider
- Anomaly (e.g. fraud) detection in a bank organization
- Person categorization, that is, identifying a human
- Air traffic control in an airport
- Streaming data from all Boeing flying air crafts

*Caution: "Prediction is very difficult, especially if it's about the future" -
Niels Bohr, Father of Quantum.*



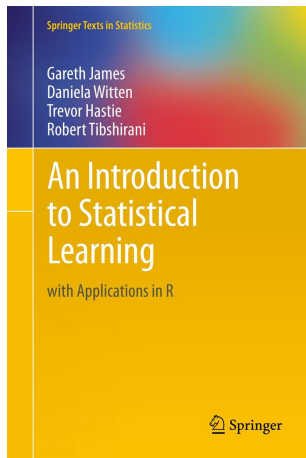


Figure: <https://www.statlearning.com/>